Aston University
BIRMINGHAM UK

DG4COM Computational Intelligence

Purpose of an assessment brief

(How should I use this brief to help with my assessment?)

This brief gives you key information and guidance about your assessment. It's the starting point for your work, and your module tutors will refer to it during assessment discussions. Read the brief carefully, along with any other materials your tutors provide, and use them to help you plan and complete your assessment.

1.0 Essential Information

Module Code and Title	DG4COM Computational Intelligence
Module/Assessment Lead contact	Prof Aniko Ekart a.ekart@aston.ac.uk
Assessment Title	ACTIVEA Physical Activity Planning Evolutionary Algorithm
Assessment weighting	20% code 30% report 50% presentation
Submission Date and Time	6 November 2026, 12:00 for report and codebase
Word Count Lengths of presentation	1700 words 8 min (+ 5 mins Q & A)

2.0 Assessment Overview

Assessment Mode (What do I need to produce at the end?)
To complete the assignment, you will need to produce:

a. **A codebase** A piece of software that you will design, implement, test and run individually. This is worth 20% of the module mark. You will import existing libraries to implement your software's core functionality, and you will write original code whilst completing the coursework tasks. Make sure your code is syntactically and semantically correct (compiles, runs, and generates a meal plan that meets the requirements set out in the tasks' description). Add comments to explain the logic implemented in every method and class. Pay attention to code design: complex logic should be divided across several simpler classes and methods. Write appropriate unit tests.

b. **A report** A 1700 word (±10%) written piece that you will prepare individually. This is worth 30% of the module mark. The report is aimed at a technical readership. Include in your report (1) the technical insights necessary for comprehending the codebase, (2) the experimental setup and results obtained whilst completing the coursework tasks, and (3) the critical analysis of the

previous two items. The critical analysis part of the report should justify your implementation and experimental design choices, explain the influence of the algorithm components (representation, genetic and selection operators, fitness function, termination criteria) on its performance (solution quality) and discuss the experimental results to support the points made throughout. Support all claims with appropriate evidence (content taught throughout the module, findings extracted from additional reading, experimental results, etc.). Pay adequate attention to the eloquence of the arguments you make, the conciseness and clarity of the narrative, the grammar, punctuation, and spelling of your report.

A presentation An 8 min (±10%) oral summary of your report and software, including a live demonstration of your software, that you will present during the week 2-6 November 2026. You will prepare a set of slides to support your presentation. You will prepare the slides and deliver the presentation individually. This is worth 50% of the module mark. The presentation is aimed at a general technical audience audience. Include in your slides diagrams and schematics covering the essentials of your code, experiments and conclusions. Accompany those with a commentary that avoids technical jargon and uses plain language to convey your findings. As part of the presentation include a live demonstration of your code. Following the presentation there will be 5 minutes for questions and answers.

Task Description (What do I have to do for this assessment?)

Design, implement and evaluate an evolutionary algorithm for evolving physical exercise sessions. An exercise session is a set of physical activities that a person performs in one workout, each with a specified duration, as illustrated below:

$session_1 = \{(jogging, 20), (bodyweight_squats, 10)\}$

$session_2 = \{(bicycling, 30), (plank, 5), (yoga, 15)\}$

Each activity in a session has a **name** and a **duration in minutes**. For example, the first session above contains 20 minutes of jogging and 10 minutes of squats, whereas the second comprises 30 minutes of cycling, 5 minutes of planking, and 15 minutes of yoga.

Each session has: a **caloric expenditure** (calories burned), and several associated **exercise intensity metrics**. The calories burned per minute and the exercise intensity metrics for each activity are stored separately, as shown below (fill in the remaining columns with appropriate values from reliable sources):

Activity	Calories/min	MET	Target RPE	Target HR [%]	Fitness domain
Jogging	7.35	6	5	70-77	CV, ME, FM

Bodyweight squats	4.9	4	6	64-70	ST, ME, BA, FM
Cycling (moderate)					
Plank					
Yoga					
Rope skipping					
Running					

Populate with appropriate data

Note that the calories/min are shown for a 70 kg adult and these would need to be modified to correspond to a person's weight.

Exercise sessions are evaluated using measurable exercise intensity metrics components:

- **Metabolic Expenditure Rate (MET):** a simple, practical, and easy way for expressing the energy cost of physical activities as a multiple of the resting metabolic rate[1].
- **Target Rate of Perceived Exertion (RPE):** the planned level of perceived effort—expressed on an established RPE scale—that an individual should aim to reach during an exercise session to achieve the intended training intensity
- **Target Heart Rate (HR):** the planned heart-rate range, expressed as a percentage of maximum heart rate that a person should reach during exercise to achieve a specific intensity level, such as the moderate zone (64–76% MHR) or vigorous zone (77–93% MHR).

These are all known exercise-science metrics and can be sourced from reputable databases and publications.

Fitness domains —such as cardiovascular fitness, strength, mobility, agility, balance and functional movement— are added to the exercise-science based metrics, as they provide a practical way to describe the different qualities an exercise develops so they help structure planning and comparison even though they are classifications rather than formal exercise-science metrics:

- **Cardiovascular Fitness (CV):** the ability to sustain activity through efficient heart-lung oxygen delivery.
- **Strength Development (ST):** the ability of muscles to generate maximal or high levels of force.
- **Muscular Endurance (ME):** the ability of muscles to perform repeated efforts or maintain contractions over time.

[1] Jetté M, Sidney K, Blümchen G. Metabolic equivalents (METS) in exercise testing, exercise prescription, and evaluation of functional capacity. Clin Cardiol. 1990 Aug;13(8):555-65. doi: 10.1002/clc.4960130809

- **Flexibility/Mobility (FL):** the ability to move joints freely through a full, usable range of motion.
- **Balance & Stability (BA):** the ability to maintain body control during static or dynamic positions.
- **Agility (AG):** the ability to change direction quickly and efficiently with coordinated body control.
- **Functional Movement (FM):** the ability to perform natural, everyday movement patterns efficiently and safely.

The core version of the algorithm suggests one ideal session, i.e., a combination of items from a given set of available activities that provides optimal exercise quality within the healthy caloric expenditure range. In other words, the algorithm's job is to find the healthiest session out of all possible combinations of activities from a given set.

What makes a session healthy, depends on the fitness level and the target of the individual. The general recommendation is to plan mixed exercise sessions for burning 300-600 calories. For more precise targets, the following recommendations are made:

- for general health 200–400 calories/session consisting of light-to-moderate mixed sessions (such as walking, strength, mobility)
- for fitness improvement, 300–500 calories/session (Moderate cardio, strength circuits) and
- for weight loss or higher-intensity training, 400–600 calories/session (such as vigorous cardio, interval training, or heavy mixed sessions).

It is also expected that a healthy session includes adequate fitness domain support.

Core tasks (up to 69 marks)

Develop a basic version of ACTIVEA that evolves the combination of activities for one session, i.e., suggests the exercise plan for one session. To do this, complete the following tasks:

1. Data setup. Compile the set of possible activities to make up the session and complete the exercise quality table for each. Separate training data from testing data.
2. Algorithm setup. Choose the type of evolutionary algorithm you will be implementing: Genetic Algorithm, Genetic Programming, Evolutionary Strategies, etc. Select the genetic operators (crossover, mutation, reproduction, etc.) and configure them to match the chosen chromosomal representation (linear, hierarchical, graph, etc.). Define the fitness function,

i.e., define a mathematical equation that captures the part of the overview explaining what makes a healthy exercise session (consult reliable exercise-science websites or hardcopy resources to identify the recommended values and ranges for MET, target RPE, target HR and fitness domains). Choose the selection operator and define the termination criteria. Make notes of the rationale behind your choices (citing supporting evidence from lecture content and background reading).

3. Algorithm implementation. Write the ACTIVEA code in accordance with the setup completed in the previous task. You may do this from scratch or by importing and customising a pre-existing library (DEAP, GPLearn, etc.). Run the core ACTIVEA algorithm and record the experimental results alongside your interpretation. Make notes of the implementation details to document the technical aspects of your code.
4. Nature-inspired optimisation. Select one nature-inspired algorithm out of those discussed in week 3, retrieve a publicly available implementation of that algorithm and run it on the same data used to train and test ACTIVEA. Compare the results against those generated by the ACTIVEA algorithm completed in the previous task. Make notes capturing your experimental insights regarding the likely causes of the differences between the exercise session recommendations produced by running the nature-inspired algorithm and ACTIVEA, respectively.
5. Parameter management. Run systematic experiments to fine-tune ACTIVEA parameters: population size, number of generations, genetic operator and selection probabilities, etc. Follow the procedure outlined in the relevant practical (week 4). Make notes of the experimental setup, results and analysis.
6. Diversity analysis. Enhance the ACTIVEA algorithm implemented in task 3 by adding one diversity preservation technique out of those discussed in week 4. Run the original version of ACTIVEA against the enhanced one and compare the results. Make notes to document your additional code and capture your experimental insights regarding the likely causes of the differences in the two exercise session recommendations.

Advanced tasks (70+ marks)

After completing all core tasks (1 through to 6), perform two of the following advanced tasks.

7. Improve the ACTIVEA algorithm by enhancing one of its genetic operators. Specifically, implement a bespoke crossover or mutation operator that improves the algorithm's chances of identifying the optimal physical activity recommendation in as computationally efficient a manner as possible. Run

systematic experiments to compare the core and enhanced versions of the ACTIVEA algorithm. Make notes to document your additional code and capture your experimental insights.

8. Improve the ACTIVEA algorithm by enhancing its fitness function to cater to exercise preferences from the seven exercise domains. Specifically, extend the fitness function formula defined in core task 2 to penalise sessions that contain activities that are not suitable for the intended preference, for example if the intended focus is CV and FM, penalise activities that do not support either CV or FM. Think carefully about the implications this has on training and testing data as well as on chromosome representation. Run systematic experiments to compare the core and enhanced versions of the ACTIVEA algorithm. Make notes to document your additional code and capture your experimental insights.
9. Improve the ACTIVEA algorithm such that it recommends several exercise sessions forming a daily, weekly, or monthly physical activity plan, depending on user preferences (these need not be provided via an elaborate interface, hardcoding them is acceptable). Run the advanced ACTIVEA algorithm and record the experimental results. Make notes to document your additional code and capture your experimental insights.
10. Continue core task 4 by selecting a second nature-inspired algorithm out of those discussed in week 3, retrieving a publicly available implementation of that algorithm and running it. Compare the results against those obtained in task 4. Make notes capturing your experimental insights regarding the likely causes of the differences between the physical activity recommendations produced by running this second nature-inspired algorithm and ACTIVEA.
11. Continue core task 6. Create a copy of the current ACTIVEA algorithm. Modify the copy by adding one diversity preservation technique out of those discussed in week 4, different to the one implemented in task 6. Run the modified copy and compare the results against those obtained in task 6. Make notes to document your additional code and capture your experimental insights.

Submission Instructions (When and Where do I submit?)

a. **Format** Your 1700 word (±10%) report should use a legible 12pt font, single spacing, and one of the standard referencing styles (Harvard, APA, Vancouver, etc.). Although there are no strict formatting restrictions regarding the presentation slides, aim to make them visually engaging, clear and concise (i.e., avoid large blocks of text and complex animations, opting instead for easy-to-understand charts, diagrams and bullet point lists).

b. **Upload** Go to your version of the **DG4COM module on Blackboard** (depending on when you started your course, the module code will have a different ending). In the **Content** tab, open the folder titled **Assessment Area**. In the **Assessment Submission** subfolder:
   - Click on the **Codebase** submission link to upload your code (.zip only, no other archive types allowed). Follow the instructions on the webpage and click Submit at the end. Do so before **noon, on 6 Nov 2026**.
   - Click on the **Report** submission link to upload your report (.docx or .pdf only, no other document formats allowed). If needed, refer to this guidance. Follow the instructions on the webpage and click Submit at the end. Do so before **noon, on 6 Nov 2026**.
   - Click on the **Presentation** submission link to upload your presentation slides (.ppt or .pdf). Follow the instructions on the webpage and click Submit at the end. Do so before **noon, on 6 Nov 2026**.

c. **Mind the rules**
   - Submit the codebase, report and presentation in the allowed document formats in order to have your work marked.
   - Submit on time. A 5% penalty per day will be applied for late submissions; submissions that are more than three days late (including weekends) will receive a mark of 0.

Get support Make sensible and timely use of university resources. If you cannot complete and submit your work on time because of exceptional circumstances, apply for an extension

3.0 Purpose of Assessment

Link to Learning Outcomes (How does the assessment help you demonstrate the outcomes of this module?)

Module Leaning Outcomes	Assessment Aspects
LO1. Implement different computational intelligence techniques, evaluate their performance and analyse the impact of different problem representations on the performance of those techniques.	Code element Experimental analysis sections in report element Demo in presentation
LO2. Judge and reason about appropriate computational intelligence techniques for a given	Critical analysis sections of report and presentation elements Code element

problem and explain the consequences of decisions. Use this understanding to design novel variants of computational intelligence techniques for solving the problem.	
LO3. Critically analyse recent research in computational intelligence.	Selection of alternative techniques to compare own algorithm against in all assessment elements

Purpose of Assessment (Why am I doing this, and what knowledge, skills or professional links should I demonstrate?)

Through this assessment you will be able to demonstrate the necessary skills to design, implement and evaluate robust computational intelligence methods for solving a real world problem, based on recent literature.

4.0 Feedback Expectations

When and where can I find my grades and feedback?

Formative Feedback

You will receive guidance on your code, report and presentation from your module tutor throughout the module. In the face-to-face sessions, the tutor will work with you to complete tasks that build skills relevant to your assessment, will give you feedback on intermediate versions of your report and presentation, will answer your specific queries and will signpost additional study resources.

Summative Feedback

You will receive comprehensive, bespoke written feedback on your code, report and presentation, on Blackboard, by 4 December 2026.

Marks

Any grades that you see outside of My Aston Portal (MAP) should be thought of as provisional. Only final Exam Board approved grades will appear in MAP. Follow this guide to access your provisional grades and feedback on Blackboard.

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5.0 Guidance on AI Usage

Can I use AI in my work and how?

Use of Gen AI is Optional 	[Example text] It is appropriate, but not essential, to make constructive use of generative AI tools for this assignment. [explain how and why] This should not include using AI to create the content of the assessment submission, but may, for example, include generating initial ideas, summarising notes or research articles, for translation, or creating a study plan. Any application or use of AI should always be acknowledged. Misuse of generative AI, as defined in Aston's assessment regulations, will be treated as an academic offence and may be subject to disciplinary action. You may find this guidance by The Learning Services team on the effective and ethical use of AI for assessment useful: Artificial Intelligence for study and assessment | Rise 360

6.0 Assessment Top Tips

What further information and advice might help me to complete this task?

- Acquire the essentials of each module section by doing the prep. Build your knowledge and skillset by actively engaging with module activities throughout taught sessions to efficiently grasp the concepts, methodologies and resources covered. Consolidate your knowledge and skillset by doing the post-teaching work on time.
- Read the assessment brief carefully, stick to the task described, follow the rules and guidance and keep to the recommended timeline.
- Incorporate formative feedback received throughout the module in your coursework. This will help you make gradual, consistent progress.
- Ask the tutor for help with your queries as soon as you think of them.
- Develop your professional comms skills with help from the University's Library and Learning Services on academic writing and giving effective presentations.

7.0 Assessment Criteria and Rubrics

What will my work be marked against?

	Codebase (20%)
Fail (0-39%)	The codebase does not compile or compiles but fails to produce a solution. No or little original code was added to the imported library. Some core tasks attempted, with significant flaws (logical mistakes in implementation, sloppy experimental work).
Borderline Fail (40 – 49%)	The codebase is functioning (runs and generates a result) with minimal original code added to the imported library. Most or all core tasks attempted, with significant flaws (logical mistakes in implementation, sloppy experimental work).
Pass (50 – 59%)	The codebase is functioning (runs and generates a result) with significant original code added to the imported library. Most or all core tasks attempted, with some flaws (logical mistakes in implementation, subpar experimental work).
Merit (60 – 69%)	The codebase is functioning and computationally efficient (takes up a reasonable amount of computational resources to generate a high-quality result) with significant original code added to the imported library. All core tasks completed, with minor or no flaws in implementation and experimental work.
Distinction (70 – 79%)	The codebase is functioning and computationally efficient (takes up a reasonable amount of computational resources to generate a high-quality result) with significant original code added to the imported library. All core tasks and one enhanced task completed, with minor or no flaws in implementation and experimental work.
Distinction + (80-100%)	The codebase is functioning and computationally efficient (takes up a reasonable amount of computational resources to generate a high-quality result) with significant original code added to the imported library. All core tasks, one enhanced task and one advanced task completed, with minor or no flaws in implementation and experimental work.

	Report (30%)
Fail (0-39%)	The report is mostly descriptive, with few or no elements of critical analysis. The argument is unconvincing, is supported by insufficient or no evidence, and lacks the

	appropriate depth. The writing has major defects: regular errors of English, substantial repetition and/or irrelevance, significant gaps in the rationale leading to flawed conclusions, poor structure, and a standard of presentation (e.g., layout, fonts, headings, figures, references) that actively detracts from understanding the message being conveyed.
Borderline Fail (40 – 49%)	The report is mostly descriptive, with some elements of critical analysis. There is an attempt at reflecting on the algorithm's implementation and on the experimental work, yet the conclusions being drawn fail to coalesce into a meaningful take-away message. The writing has some defects that do not detract from understanding the message being conveyed.
Pass (50 – 59%)	There is a good balance between the descriptive and the analytical/reflective elements of the report. The arguments made are clearly articulated and accompanied by adequate evidence. The report is clearly structured and makes effective use of figures and other supporting artefacts. Any writing defects are minor.
Merit (60 – 69%)	There is a good balance between the descriptive and the analytical/reflective elements of the report. The arguments made are clearly articulated and accompanied by extensive evidence. The report is clearly structured and makes effective use of diagrams and other supporting artefacts. Any writing defects are minor.
Distinction (70 – 79%)	There is a remarkably well-calibrated balance between the descriptive and the analytical/reflective elements of the report. The arguments made are clear, concise, and accompanied by particularly significant evidence, which has been carefully selected from an extensive body of literature. The report is structured in a highly intuitive manner and makes effective use of diagrams and other supporting artefacts. Any writing defects are very minor.
Distinction + (80-100%)	There is a remarkably well-calibrated balance between the descriptive and the analytical/reflective elements of the report. The arguments made are clear, concise, and feature elements of innovation. The narrative draws from an extensive and diverse pool of particularly significant evidence. The report is structured in a highly intuitive

	manner and makes excellent use of diagrams and other supporting artefacts. Any writing defects are very minor.

	Presentation (50%)
Fail (0-49%)	The presentation is superficial, incomplete, and delivered in a hesitant, unconvincing, and/or incoherent manner. The student could not answer questions about the presentation.
Borderline Fail (40 – 49%)	The presentation and answers to questions provide limited insight, are incomplete, and delivered in a hesitant, unconvincing, and/or incoherent manner.
Pass (50 – 59%)	The presentation and answers to questions demonstrate an adequate level of insight, are complete, and delivered in a confident, convincing, and coherent manner.
Merit (60 – 69%)	The presentation and answers to questions provide substantial insight, are complete, and delivered in a confident, convincing, and coherent manner.
Distinction (70 – 79%)	The presentation and questions provide substantial insight, demonstrate a remarkable level of critical analysis, are complete, and delivered in a confident, convincing, and coherent manner.
Distinction + (80-100%)	The presentation and answers to questions are elegant, provide substantial insight, demonstrate a remarkable level of critical analysis, are complete and delivered in a confident, convincing, and coherent manner that can be described as highly engaging, innovative, and/or illuminating.